ABSTRACT

The Hospital Management System is a full-stack web application developed to modernize, automate, and efficiently manage the day-to-day operations of a hospital. The system provides a centralized digital platform that enables seamless interaction between different users such as administrators, doctors, receptionists, patients, and billing staff. It supports various functionalities including patient registration, appointment scheduling, medical record management, billing, and real-time data access, thereby improving the overall workflow of the hospital.

The application is developed using modern web technologies to ensure scalability, performance, and ease of use. The frontend is designed using HTML, CSS, and JavaScript to create a responsive and user-friendly interface that enhances user experience across different devices. The backend is implemented using Node.js, which efficiently handles server-side logic and ensures smooth communication between the client and the database. MongoDB is used as the database system, providing a flexible and scalable solution for storing and managing large volumes of hospital data in a structured manner.

One of the primary objectives of this system is to reduce manual effort and eliminate inefficiencies present in traditional hospital management methods. In many hospitals, data is still maintained manually or through disconnected systems, leading to errors, redundancy, and delays. This system addresses these challenges by automating critical processes such as patient data management, appointment booking, and billing operations. It minimizes human errors, improves accuracy, and significantly reduces the time required for administrative tasks.

The system also incorporates role-based access control, ensuring that each user can access only the functionalities relevant to their role. This enhances data security and maintains confidentiality of sensitive patient information. Real-time data updates and centralized storage further improve communication and coordination between departments, enabling faster decision-making and better patient care.

In addition, the system is designed with scalability and flexibility in mind, allowing it to handle increasing numbers of users and data as the hospital grows. Its modular architecture makes it easy to upgrade and maintain, ensuring long-term usability.

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1. INTRODUCTION

The Hospital Management System (HMS) is a web-based application developed to efficiently manage and automate the daily operations of a hospital. It provides a centralized platform for handling patient records, appointment scheduling, doctor management, and billing processes in an organized and secure manner.

In traditional hospital systems, most operations are carried out manually using paper-based records. This leads to several issues such as data redundancy, human errors, difficulty in retrieving information, and delays in providing services. Managing large volumes of patient data manually becomes inefficient and time-consuming. To overcome these challenges, a digital and automated system is essential.

This project introduces a role-based Hospital Management System where different users such as Administrator, Receptionist, Doctor, Patient, and Billing Staff have controlled access based on their responsibilities. Each role performs specific tasks, ensuring smooth workflow and better coordination among hospital departments.

The system is developed using modern web technologies such as HTML, CSS, and JavaScript for the frontend, with a backend environment connected to a MongoDB database for efficient data storage and management. MongoDB, being a NoSQL database, allows flexible and scalable handling of large amounts of data.

The main objective of this system is to reduce manual workload, improve data accuracy, and provide faster access to information. Overall, the Hospital Management System enhances operational efficiency and contributes to better healthcare service delivery.

1.1 OBJECTIVE

The main objective of this project is to design and develop a centralized Hospital Management System that efficiently manages and streamlines various hospital operations. The system aims to replace traditional manual methods with a digital platform that improves accuracy, speed, and accessibility of information.

The project focuses on integrating different hospital functionalities into a single system, allowing smooth coordination between departments such as reception, doctors, and billing. It also ensures that data is stored securely and can be accessed easily by authorized users.

Specific Objectives:

- To develop a secure role-based access system for different users
- To maintain and manage digital records of patients efficiently
- To simplify and automate the appointment scheduling process
- To implement an automated billing system for accurate calculations
- To ensure secure, reliable, and efficient data handling
- To reduce manual workload and minimize human errors
- To improve communication and coordination between departments

1.2 MOTIVATION

The motivation behind developing this system arises from the limitations and challenges faced in traditional hospital management systems. Many hospitals still rely on manual or partially computerized methods, which lead to inefficiencies and errors in managing patient and administrative data.

Manual record keeping often results in data loss, duplication, and difficulty in maintaining large volumes of information. Administrative processes such as appointment booking and billing are time-consuming and require significant human effort. Additionally, there is often a lack of proper communication between hospital staff, which affects the overall workflow and patient experience.

Accessing patient records in traditional systems can also be difficult and time-consuming, especially during emergencies. These challenges highlight the need for a more efficient and reliable solution.

This project is motivated by the need to overcome these issues by developing a smart, user-friendly, and efficient system using modern web technologies. The aim is to improve hospital operations, enhance data management, and provide better services to patients through digital transformation.

2. COMPANY PROFILE



ORMATRIX – The Complete IT Solution

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About the Company

ORMATRIX – Complete IT Solutions is a growing software development company that focuses on delivering innovative and efficient digital solutions. The company specializes in developing web-based applications, software systems, and mobile applications tailored to business needs.

ORMATRIX aims to provide high-quality IT services using modern technologies and industry best practices. The company works with clients from different domains and helps them digitize their processes to improve productivity and efficiency.

Vision and Mission

Vision:

To become a leading IT solutions provider by delivering innovative, scalable, and reliable software products.

Mission:

- To provide high-quality and cost-effective IT services
- To adopt modern technologies for better solutions
- To ensure customer satisfaction through efficient service delivery
- To support digital transformation for businesses

Services Offered

ORMATRIX provides a wide range of IT services, including:

- **Web Application Development**
  Development of dynamic and responsive web applications
- **Software Development**
  Customized software solutions for business automation
- **Mobile App Development**
  Android and cross-platform application development
- **UI/UX Design**
  Designing user-friendly interfaces and experiences
- **Database Management**
  Efficient data storage solutions using databases like MongoDB
- **IT Consulting**
  Providing technical guidance and project planning

Technologies Used

The company works with modern technologies such as:

- HTML, CSS, JavaScript
- MongoDB (NoSQL Database)
- Node.js (for backend development)
- REST APIs
- Cloud-based tools

Work Environment

ORMATRIX provides a professional and learning-oriented work environment. Interns and employees are encouraged to work on real-time projects, which helps in gaining practical knowledge and industry exposure.

The company promotes teamwork, innovation, and continuous learning.

Internship Experience

During my internship at ORMATRIX, I worked on the **Hospital Management System** project. This project helped me understand:

- Real-time application development
- Role-based system implementation
- Database design using MongoDB
- Frontend and backend integration
- Testing and debugging

The internship provided valuable hands-on experience and improved my technical and problem-solving skills.

3. DAILY WORK REPORT

Day 1

- Joined the internship at ORMATRIX – Complete IT Solutions
- Introduction to company environment and workflow
- Overview of Hospital Management System project

Day 2

- Understood project requirements and objectives
- Studied basic hospital workflow and operations

Day 3

- Identified system modules: Admin, Doctor, Receptionist, Patient, Billing
- Learned about role-based access system

Day 4

- Introduction to MongoDB database
- Learned about collections and document structure

Day 5

- Designed database schema for patients and doctors
- Planned data relationships

Day 6

- Created basic UI layout using HTML
- Designed login page structure

Day 7

- Styled login page using CSS
- Added form validation using JavaScript

Day 8

- Developed dashboard UI for Admin
- Created navigation menu

Day 9

- Developed receptionist interface
- Designed patient registration form

Day 10

- Implemented patient data entry system
- Connected frontend with backend

Day 11

- Developed doctor dashboard
- Added functionality to view appointments

Day 12

- Created appointment booking module
- Designed appointment scheduling interface

Day 13

- Integrated MongoDB database with application
- Stored patient and appointment details

Day 14

- Implemented role-based login system
- Restricted access for different users

Day 15

- Developed patient management module
- Added update and delete options

Day 16

- Created billing module interface
- Designed bill generation format

Day 17

- Implemented billing logic
- Stored billing records in database

Day 18

- Integrated all modules together
- Ensured smooth navigation between pages

Day 19

- Performed unit testing of modules
- Checked functionality of each role

Day 20

- Debugged errors in database queries
- Improved system performance

Day 21

- Tested appointment scheduling system
- Verified data accuracy

Day 22

- Improved user interface design
- Enhanced responsiveness

Day 23

- Added validations to forms
- Prevented incorrect data entry

Day 24

- Conducted system testing
- Checked workflow from login to billing

Day 25

- Fixed remaining bugs and errors
- Optimized application performance

Day 26

- Final integration testing
- Verified all modules working correctly

Day 27

- Reviewed project with supervisor
- Made necessary corrections

Day 28

- Prepared project documentation
- Organized project files

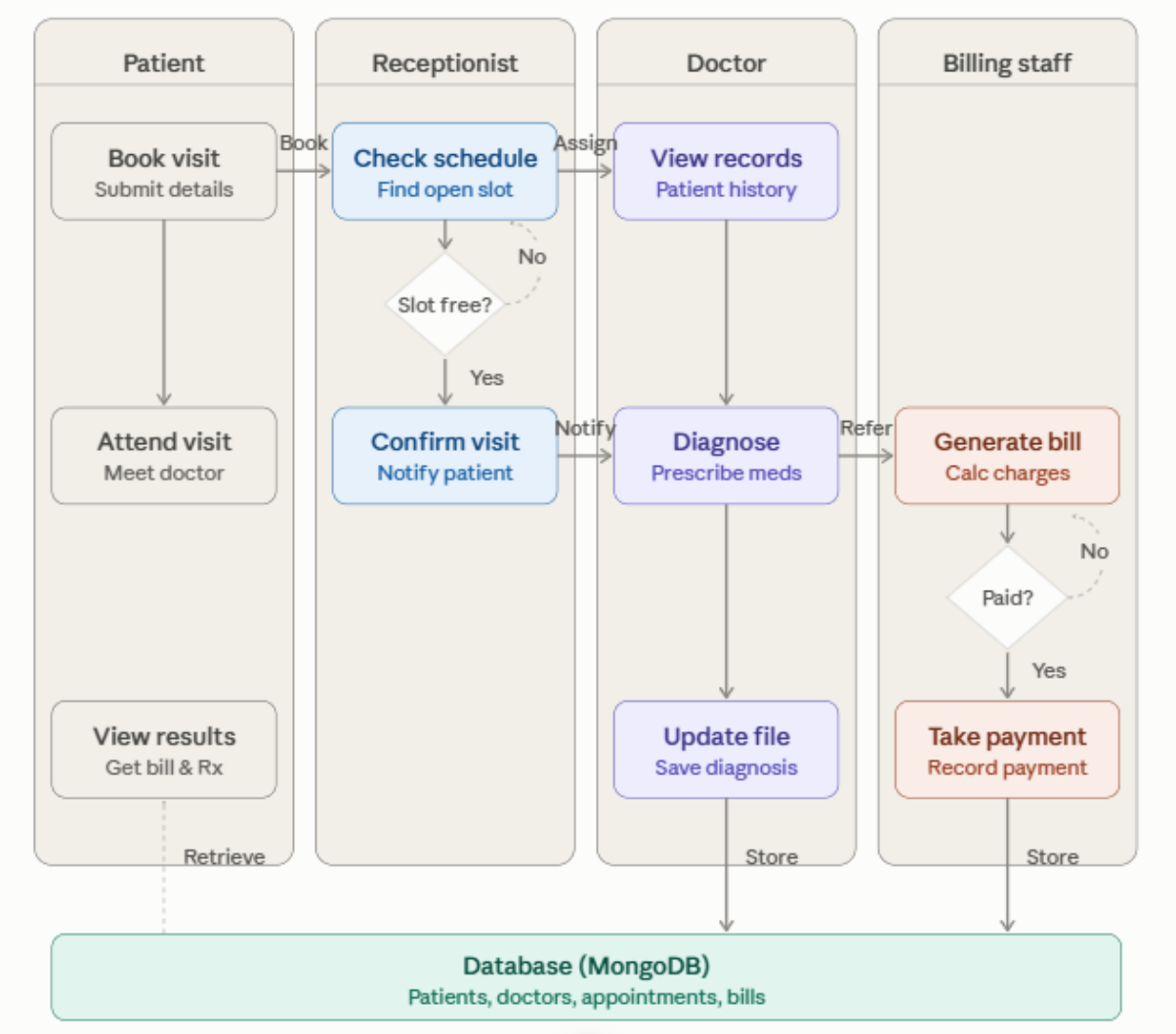
Day 29

- Final testing and validation
- Ensured system stability

Day 30

- Completed project submission
- Gained overall understanding of system

4.BLOCK DIAGRAM



5. HOSPITAL MANAGEMENT SYSTEM

The Hospital Management System is a web-based application developed during the internship to manage and streamline hospital operations efficiently. The system integrates various modules such as patient management, doctor management, appointment scheduling, and billing into a single platform.

This system is designed with **role-based access control**, allowing different users like Administrator, Receptionist, Doctor, Patient, and Billing Staff to access only the functionalities

assigned to them. The application uses modern web technologies with **MongoDB as the database**, ensuring flexibility and scalability.

The main objective of this system is to reduce manual work, improve accuracy, and enhance overall hospital management.

5.1 EXISTING SYSTEM

In many hospitals, traditional methods are still used for managing data. These systems are either manual or partially computerized, which leads to inefficiency in handling daily operations. Patient records are usually maintained on paper, and tasks like appointment booking and billing are performed manually.

This approach makes the system slow and difficult to manage, especially when handling a large number of patients. There is no proper integration between departments, which affects communication and overall workflow.

Features of Existing System:

- Patient records maintained on paper
- Manual appointment booking
- Billing done manually or using basic tools
- No centralized data management
- Limited or no role-based access

Disadvantages of Existing System:

1. Time Consuming

Manual data entry and searching for records take a lot of time.

2. Error-Prone

High chances of human errors in maintaining records and billing.

3. Data Redundancy

Duplicate entries and inconsistent data may occur.

4. Lack of Security

Patient data is not stored securely.

5. Poor Accessibility

Data cannot be accessed easily across departments.

6. Inefficient Workflow

Poor coordination between departments leads to delays.

5.2 PROPOSED SYSTEM

The proposed system is a web-based Hospital Management System that automates all major hospital operations. It provides a centralized platform where all data is stored and managed efficiently.

The system is developed using HTML, CSS, and JavaScript for the frontend, Node.js for backend processing, and MongoDB as the database. It includes role-based access so that each user can access only the required features.

This system improves efficiency, reduces manual work, and ensures accurate and secure data handling.

Features of Proposed System:

- Role-based login system
- Centralized database using MongoDB
- Automated patient registration
- Online appointment scheduling
- Digital billing system
- Real-time data access

Advantages of Proposed System:

1. Automation

Reduces manual work and improves efficiency.

2. Accuracy

Minimizes errors in data and billing.

3. Data Security

Provides secure login and controlled access.

4. Centralized System

All data is stored in one place.

5. Easy Accessibility

Authorized users can access data anytime.

6. Scalability

Can handle increasing data and users.

5.3 CAPABILITIES

The system provides multiple capabilities to improve hospital operations. It supports role-based access, allowing administrators, receptionists, doctors, patients, and billing staff to perform their respective tasks efficiently.

It includes patient management features such as adding, updating, viewing, and deleting patient records. Appointment scheduling allows booking and managing doctor appointments while avoiding conflicts.

The billing system automatically generates bills and stores records for future use. Data is stored in MongoDB, ensuring fast and reliable access.

The system also offers a user-friendly interface with simple navigation and real-time updates, helping improve coordination and decision-making within the hospital.

6. PROPOSED METHOD

The proposed method describes the structured approach followed to design and develop the Hospital Management System. The system is implemented as a full-stack web application using modern technologies with MongoDB as the database. The development follows a modular approach, where each functionality is developed separately and then integrated into

a complete working system. This ensures better organization, easier debugging, and improved scalability.

The system mainly focuses on automating hospital operations such as patient management, appointment scheduling, and billing. It reduces manual effort and improves efficiency by providing a centralized digital platform. The use of role-based access ensures that each user interacts only with the features relevant to their responsibilities.

6.1 ROLE-BASED ACCESS

The system is designed using Role-Based Access Control (RBAC), which ensures that users can only access the functionalities assigned to their role. This improves system security and prevents unauthorized access to sensitive data.

Administrator

The Administrator has complete control over the system. This role is responsible for managing all users, including doctors, receptionists, patients, and billing staff. The administrator can add new users, update existing user details, and remove users when necessary. In addition, the administrator monitors overall system activity, ensures smooth functioning of all modules, and handles system-level configurations. This role acts as the central authority of the system.

Receptionist

The Receptionist plays a key role in handling patient interactions. This role is responsible for registering new patients by entering their personal and medical details into the system. The receptionist also manages appointment scheduling by assigning available time slots to patients based on doctor availability. In addition, the receptionist can update patient details and manage daily appointment records. This reduces manual paperwork and ensures accurate data entry.

Doctor

The Doctor accesses the system to view patient information and manage medical records. Doctors can view their scheduled appointments and access patient history before consultation. After examining the patient, doctors can update diagnosis details, prescriptions, and treatment information in the system. This ensures that all patient records are maintained digitally and can be accessed anytime when needed.

Patient

The Patient role provides limited access to personal information. Patients can log in to view their appointment details, doctor information, and medical records. This improves transparency and allows patients to stay informed about their treatment and schedules without depending on manual communication.

Billing Staff

The Billing Staff is responsible for managing all financial transactions in the system. This role generates bills for patients by adding consultation fees, medical charges, and other service costs. The billing staff ensures that all calculations are accurate and that billing records are stored properly in the database. This reduces errors and maintains a clear financial record for the hospital.

6.2 PATIENT MANAGEMENT

The patient management module is responsible for handling all patient-related information within the system. It allows the receptionist to register new patients by entering essential details such as name, age, gender, contact information, and medical history. Once the data is entered, it is validated and stored in the MongoDB database. The system allows updating and deleting patient records whenever necessary, ensuring that the data remains accurate and up to date. Doctors can access patient details during consultation, which helps in providing better treatment. This module eliminates the need for paper records and enables quick retrieval of patient information.

6.3 APPOINTMENT SCHEDULING

The appointment scheduling module is designed to manage the booking and organization of appointments between patients and doctors. The receptionist selects the appropriate doctor and assigns a suitable time slot based on availability. The system checks for conflicts and prevents double booking, ensuring efficient time management. Once the appointment is confirmed, the details are stored in the database and can be accessed by both the doctor and the patient. This module helps in reducing waiting time, improving hospital workflow, and maintaining a well-organized schedule for daily operations.

6.4 BILLING MANAGEMENT

The billing management module handles all financial aspects of the hospital system. It allows the billing staff to generate bills by adding various charges such as consultation fees, treatment costs, laboratory charges, and medication expenses. The system automatically calculates the total amount, reducing manual errors. Once generated, the bill is stored in the database for future reference. This module ensures transparency in billing, simplifies financial management, and provides accurate records for both the hospital and patients.

6.5 DATABASE MANAGEMENT

The system uses MongoDB as the database to store and manage all information efficiently. MongoDB is a NoSQL database that stores data in a flexible, document-oriented format, making it suitable for handling large volumes of data. All system data, including user details, patient records, appointments, and billing information, is stored in separate collections. This allows fast data retrieval and efficient storage. The database design ensures that data is organized, secure, and easily accessible whenever required.

6.6 SYSTEM INTEGRATION

All modules of the Hospital Management System are integrated to function as a single unified system. When a user logs in, the system identifies the role and provides access to the corresponding dashboard. Each module interacts with the database to retrieve and update

data in real time. For example, patient registration data is used in appointment scheduling and billing modules. This integration ensures smooth data flow and consistent information across all functionalities.

6.7 DEVELOPMENT APPROACH

The development of the system follows a modular and incremental approach. Each module is developed independently and tested to ensure proper functionality. After successful testing, the modules are integrated step by step to form the complete system. This approach helps in identifying errors early, simplifies debugging, and improves the overall quality of the application. Continuous testing and refinement are carried out to ensure that the system meets all functional requirements and performs efficiently.

7. ALGORITHMS

7.1 USER AUTHENTICATION

1. Start
2. User enters username and password
3. Validate input fields (check for empty values)
4. Send login request to server
5. Retrieve user data from MongoDB database
6. Compare entered credentials with stored data
7. If credentials match:
    - Identify user role (Admin/Doctor/Receptionist/Patient/Billing Staff)
    - Redirect to respective dashboard
8. Else:
    - Display error message “Invalid Credentials”
9. End

7.2 APPOINTMENT BOOKING PROCESS

1. Start
2. Receptionist selects patient
3. Choose doctor and preferred date/time
4. Check database for doctor availability
5. If slot is available:
    - Confirm appointment
    - Store appointment details in database
6. Else:
    - Display “Slot Not Available”
    - Suggest alternative time
7. Update appointment list
8. End

7.3 BILL GENERATION LOGIC

1. Start
2. Select patient details
3. Add consultation fee
4. Add treatment/lab charges
5. Add medication costs
6. Calculate total amount
7. Generate bill
8. Store bill details in database
9. Display bill to user
10. End

7.4 ROLE-BASED ACCESS CONTROL

1. Start
2. User logs in
3. Retrieve role from database
4. Match role with permissions
5. If role = Admin → Full access
6. If role = Receptionist → Patient & Appointment access
7. If role = Doctor → Patient & diagnosis access
8. If role = Patient → View-only access
9. If role = Billing Staff → Billing access
10. Restrict unauthorized actions
11. End

7.5 DATA STORAGE AND RETRIEVAL

1. Start
2. Receive request (Insert / Update / Delete / Fetch)
3. Connect to MongoDB database
4. Select appropriate collection
5. Perform requested operation
6. If operation successful:
    - Return success response
7. Else:
    - Return error message
8. Close database connection
9. End

7.6 REAL-TIME DASHBOARD UPDATES

1. Start
2. User logs into system
3. Fetch latest data from database
4. Display data on dashboard
5. Monitor for changes (new appointments, updates)
6. Refresh dashboard dynamically
7. Show updated information instantly
8. End

8. CODE IMPLEMENTATION

8.1 USER AUTHENTICATION MODULE

```
app.post('/login', async (req, res) => {
   const { username, password } = req.body;
   const user = await User.findOne({ username: username });
   if (!user) {
      return res.send("User not found");
   }
   if (user.password === password) {
      res.send({ message: "Login successful", role: user.role });
   } else {
      res.send("Invalid password");
   }
});
```

8.2 PATIENT MANAGEMENT MODULE

```
app.post('/addPatient', async (req, res) => {
   const patient = new Patient(req.body);
   await patient.save();
   res.send("Patient added successfully");
});
app.get('/patients', async (req, res) => {
   const patients = await Patient.find();
   res.json(patients);
});
```

8.3 APPOINTMENT SCHEDULING MODULE

```
app.post('/appointment', async (req, res) => {
   const { doctorId, date, time } = req.body;
   const existing = await Appointment.findOne({ doctorId, date, time });
   if (existing) {
      return res.send("Slot not available");
   }
   const appointment = new Appointment(req.body);
   await appointment.save();
   res.send("Appointment booked successfully");
});
```

8.4 BILLING MODULE

```
app.post('/billing', async (req, res) => {
    const { consultation, lab, medicine } = req.body;
    const total = consultation + lab + medicine;
    const bill = new Billing({
        ...req.body,
        totalAmount: total
    });
    await bill.save();
    res.json({ message: "Bill generated", total });
});
```

8.5 ROLE-BASED ACCESS (MIDDLEWARE)

```
function authorize(role) {
    return (req, res, next) => {
        if (req.user.role !== role) {
            return res.status(403).send("Access denied");
        }
        next();
    };
}
```

8.6 MONGODB CONNECTION

```
const mongoose = require('mongoose');

mongoose.connect('mongodb://127.0.0.1:27017/hospitalDB', {
   useNewUrlParser: true,
   useUnifiedTopology: true
})
.then(() => console.log("MongoDB Connected"))
.catch(err => console.log(err));
```

8.7 SAMPLE SCHEMA (PATIENT)

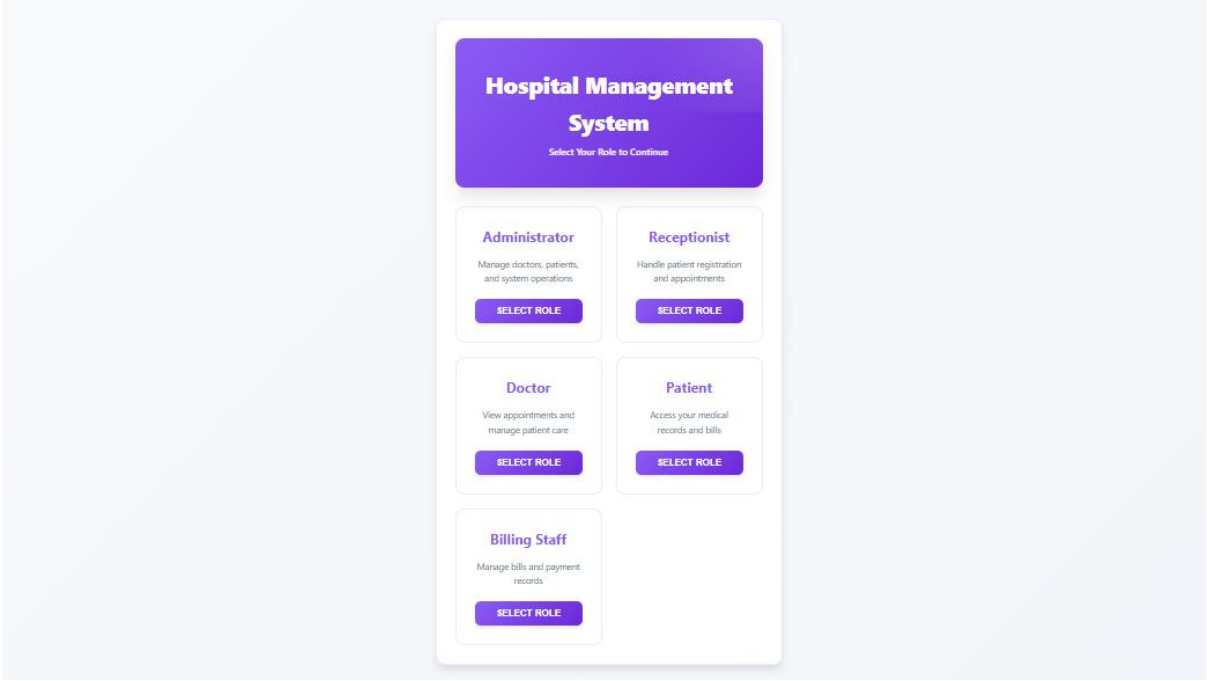
```
const mongoose = require('mongoose');

const patientSchema = new mongoose.Schema({
   name: String,
   age: Number,
   gender: String,
   contact: String,
   history: String
});

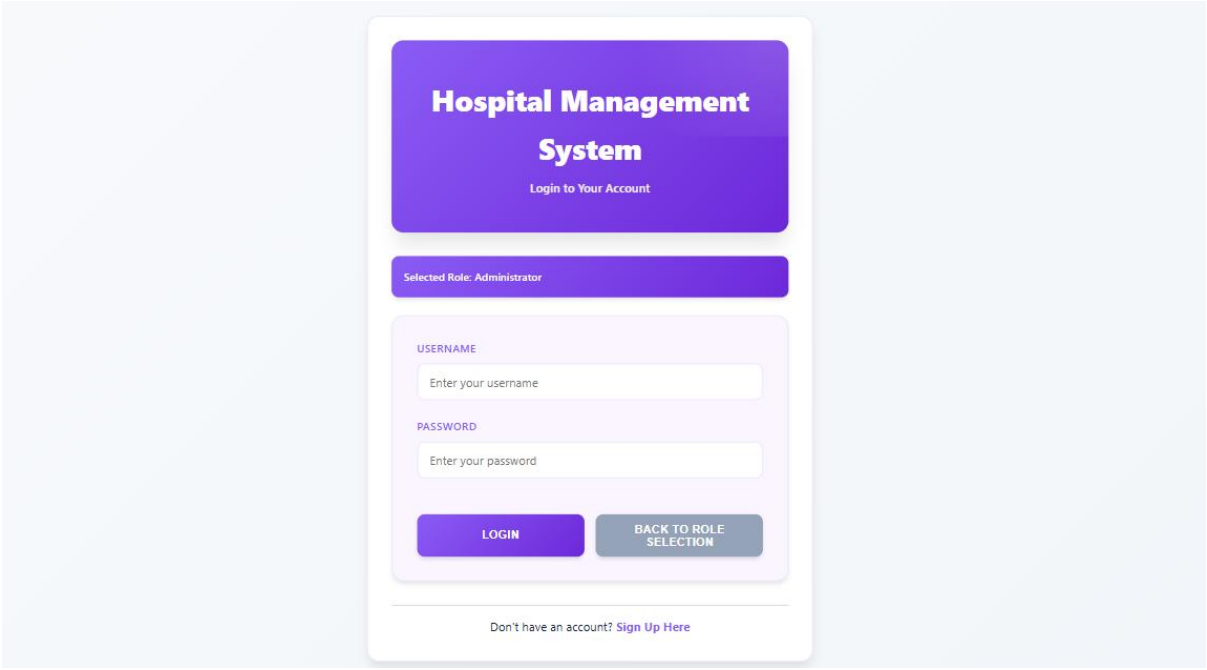
module.exports = mongoose.model('Patient', patientSchema);
```

9.SAMPLE OUTPUT

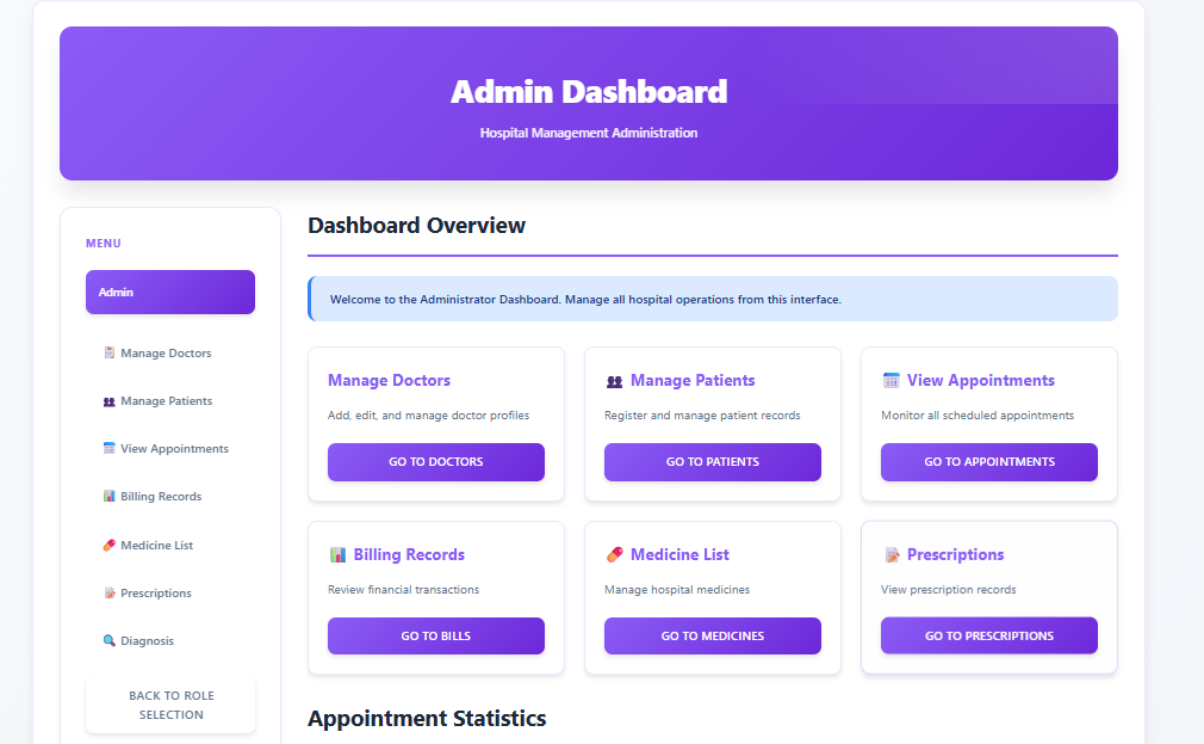
9.1 ROLE SELECTION PAGE

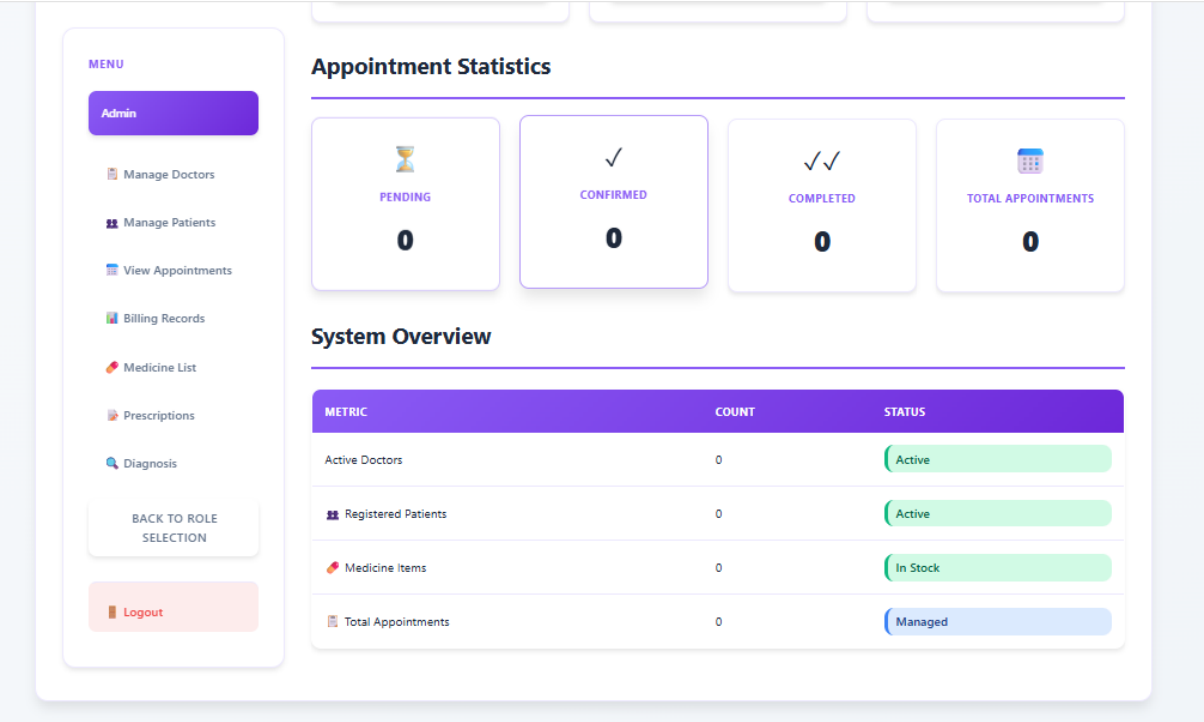


9.2 LOGIN PAGE

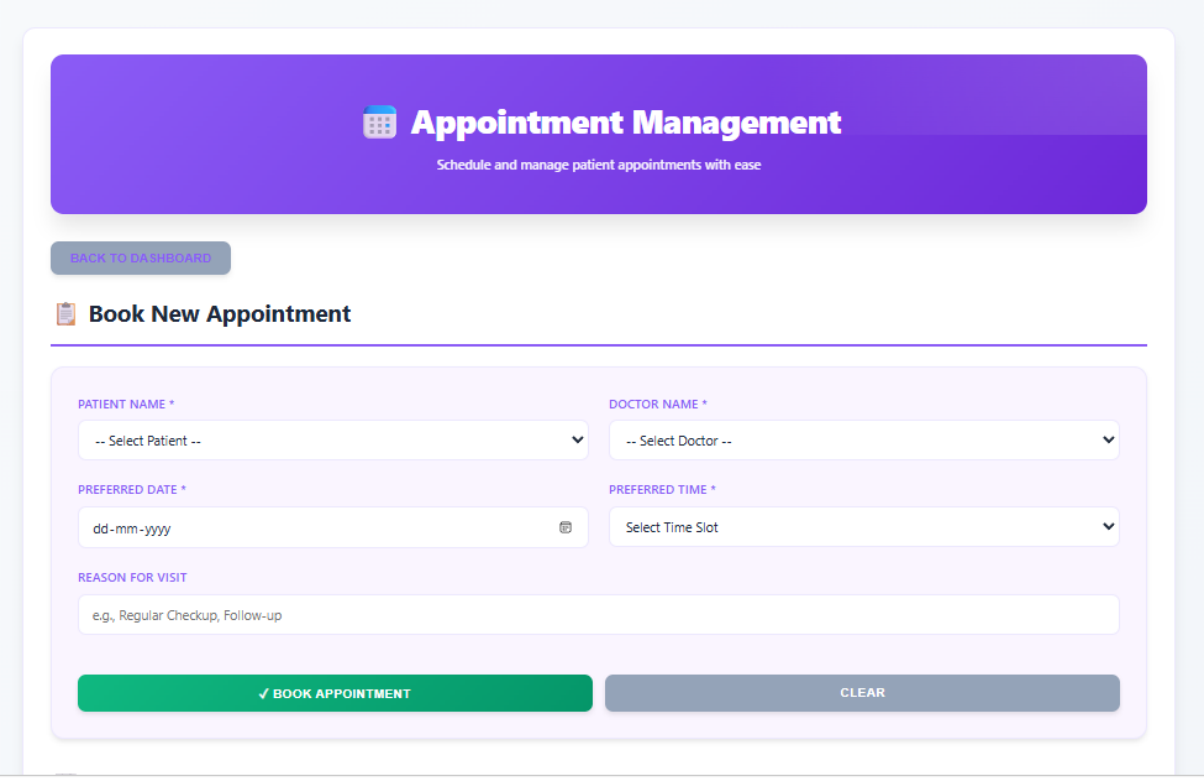


9.3 ADMIN DASHBOARD PAGE

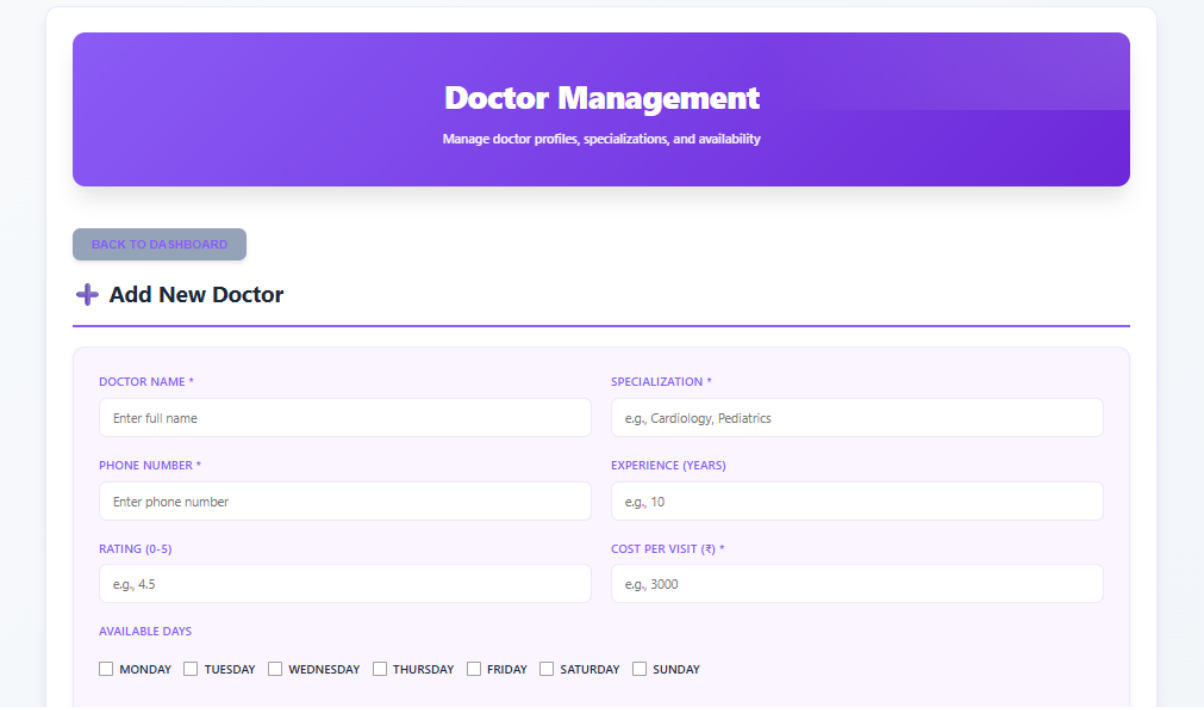




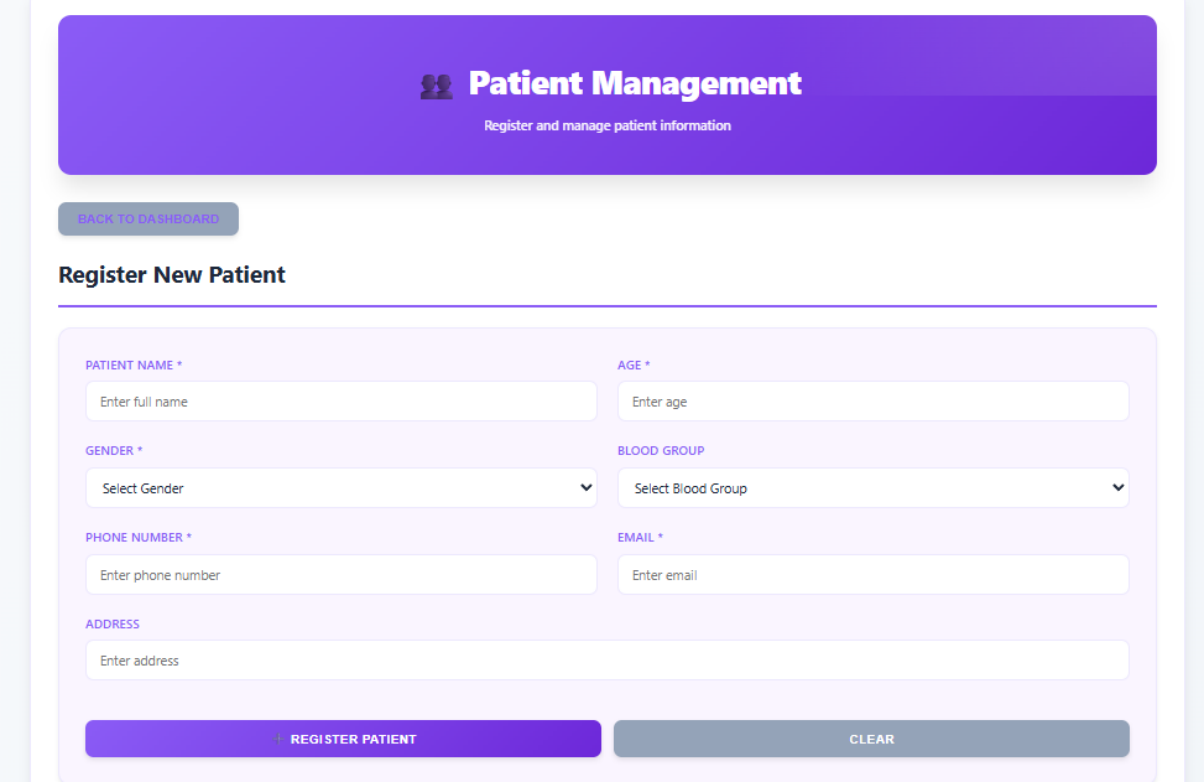
9.4 APPPOINTMENT PAGE



9.5 DOCTOR DASHBOARD PAGE



9.6 PATIENT DASHBOARD PAGE



10. ANALYSIS

Testing of the Hospital Management System was carried out module by module to validate both functional correctness and overall system behavior. Functional analysis confirmed that core operations such as user authentication, patient registration, appointment scheduling, billing generation, and role-based access control performed accurately under both normal and edge-case conditions, including invalid login attempts, duplicate entries, incomplete form submissions, and scheduling conflicts. The system effectively handled validation at both frontend and backend levels, ensuring reliable data processing.

Performance analysis indicated that the system responds quickly to user requests due to the efficient handling of operations through Node.js and the use of MongoDB as a NoSQL database. Data retrieval and storage operations such as fetching patient records, booking appointments, and generating billing details were executed with minimal latency. The system

is designed to scale efficiently as data volume increases, making it suitable for real-world applications in small to medium-sized hospitals.

Usability analysis showed that the system provides a simple and intuitive interface. The web-based design ensures accessibility across different devices, and the structured dashboards for Administrator, Doctor, Receptionist, Patient, and Billing Staff reduce complexity in performing tasks. Compared to traditional manual systems, users can complete operations such as patient registration or appointment booking in significantly fewer steps.

Comparative analysis with the existing manual system highlights major improvements. The system eliminates paperwork, reduces human errors, and ensures centralized data storage. Appointment scheduling is more organized, billing calculations are accurate, and real-time data access improves decision-making. Administrators gain better control and visibility over hospital operations, which was not possible in the traditional system.

Overall, the analysis confirms that the Hospital Management System successfully meets its objectives of improving efficiency, ensuring data accuracy, enhancing security, and providing a user-friendly platform for managing hospital operations.

11. CONCLUSION

The Hospital Management System successfully demonstrates how modern full-stack web technologies can be effectively utilized to automate and enhance hospital operations. By replacing traditional manual and paper-based methods with a centralized digital platform, the system significantly reduces administrative workload, minimizes human errors, and improves the overall efficiency and accuracy of hospital management processes.

The system integrates multiple essential modules such as user authentication, patient management, appointment scheduling, and billing into a single, well-organized platform. This integration ensures smooth coordination between different departments and enables seamless data flow across the system. The implementation of role-based access control further enhances security by restricting access to authorized users, while also improving usability by providing role-specific functionalities.

The use of MongoDB as the database ensures flexible, scalable, and efficient data storage, allowing the system to handle large volumes of patient and operational data. Node.js provides a robust backend environment that supports efficient processing of requests and smooth interaction between the frontend and database. The frontend design, built using HTML, CSS, and JavaScript, offers a user-friendly and responsive interface that enhances user experience.

Throughout the development process, all modules were carefully designed, implemented, and tested to ensure proper functionality and reliability. The successful integration and testing of the system confirm its practical applicability in real-world hospital environments. The project also provided valuable hands-on experience in various aspects of software development, including full-stack development, database management, API integration, debugging, and system testing.

In addition to meeting its functional requirements, the system improves communication between hospital staff, enables faster decision-making, and ensures better management of patient information. It also lays a strong foundation for future enhancements such as cloud integration, mobile application support, and advanced analytics.

Overall, the Hospital Management System is a reliable, efficient, and scalable solution that demonstrates the importance of digital transformation in the healthcare sector. It highlights how technology can be leveraged to improve service quality, optimize operations, and enhance patient satisfaction, making it a valuable contribution toward modern healthcare management.

12. FUTURE WORK

- Integration with advanced authentication mechanisms such as JWT for improved security
- Implementation of multi-user login with enhanced password encryption
- Development of a mobile application for Android and iOS platforms
- Integration of real-time notifications for appointments and billing updates
- Cloud deployment (AWS, Azure, or GCP) for better scalability and accessibility
- Addition of medical report upload and management system
- Implementation of analytics dashboards for hospital performance monitoring
- Integration of online payment systems for billing transactions

- AI-based patient management features such as predictive analysis and recommendations
- Enhancement of UI/UX for improved user experience

13. REFERENCES

1. W3Schools – HTML, CSS, JavaScript learning resources
2. MDN Web Docs – Web development documentation and references
3. GeeksforGeeks – Node.js, MongoDB, and full-stack development concepts
4. Node.js Official Documentation – Backend development and server-side programming
5. MongoDB Official Documentation – Database design and operations
6. Express.js Documentation – API development and routing
7. Stack Overflow – Programming discussions and problem-solving